

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Uses a limited training dataset	May affect prediction accuracy for new data	High
2	Local deployment only	Limited accessibility for users	Medium
3	Single machine processing	Reduced scalability for large workloads	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on cloud platforms such as AWS, Azure, or Render
2	Data Storage	Local dataset storage	Integrate cloud databases such as PostgreSQL or MongoDB
3	Performance	Basic Linear Regression model	Optimize model and implement caching mechanisms
4	Security	Basic input validation	Add authentication, authorization, and HTTPS support

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve dataset quality and model accuracy	3 Months	More accurate HDI predictions
Phase 2	Deploy application on cloud	6 Months	Global accessibility
Phase 3	Add interactive dashboards and visualizations	9 Months	Better user experience
Phase 4	Integrate real-time development data APIs	12 Months	Up-to-date predictions and analytics

Scalability & Future Plan Summary

The current HDI Predictor successfully predicts Human Development Index values and classifies countries into development categories. To support future growth, the system can be deployed on cloud infrastructure, integrated with scalable databases, and enhanced with advanced machine learning algorithms. Future development will focus on improving prediction accuracy, adding real-time data integration, strengthening security, and providing advanced analytical dashboards for policymakers and researchers.